(*iii*) **Gamma** (γ) **rays** (no charge and no mass) They are not deflected from their path in the electric or magnetic field. These are electromagnetic radiations and have very high penetrating power. Emission of γ-rays is the secondary effect of radioactive charge.

Note

1. Stable nuclei are those for which number of neutrons and protons are equal.
2. The time taken by half of the atoms of a radioactive element to disintegrate is called its **half-life**. Its unit is time^{-1}.
3. X-rays were discovered by Roentgen in 1896. These are electromagnetic waves of very short wavelength and are used to detect cracks in fractured bones.
4. If the energies of α, β and γ-particles is same, then penerating power $\alpha < \beta < \gamma$.

Group Displacement Law

This law was put forward by Soddy, Fajan and Russel on the basis of following fact :

(*i*) When a radioactive element emits an α-particle, the atomic number of the resulting nuclide decreases by 4 units e.g.

$$^{238}_{92}\text{U} \longrightarrow \,^{234}_{90}\text{Th} + \,^{4}_{2}\text{He}$$

(*ii*) When a radioactive element emits a β-particle, the atomic number of the resulting nuclide increases by one unit but the atomic mass remains unchanged e.g.,

$$^{238}_{92}U \longrightarrow \,^{238}_{93}Np + \,^{0}_{-1}e$$

NUCLEAR REACTIONS

These are of following two types:

Nuclear Fission

The splitting of a heavy nucleus into two smaller of nearly comparable masses and release of about 200MeV of energy is called nuclear fission. Nuclear fission was discovered by 'Hahn' and 'Strassmann' in 1939.

- Atom bomb is the result of uncontrolled nuclear fission.
- The device in which controlled nuclear fission (chain reactions) is carried out is called **nuclear reactor**. The fission is controlled by absorbing neutrons by using cadmium or boron rods.
- Heavy water (D_2O, molecular weight 20) and graphite are used as moderator for slowing down the fast moving neutrons. U^{235} is used as a nuclear fuel.

Nuclear Fusion

The union of (two or more) lighter nuclei to form a heavier nucleus is called the nuclear fusion. It is also accompanied by release of energy because the total mass of products is lesser than the total mass of reactants.

- Nuclear fusion occurs only at extremely high temperature ($> 10^6$K), so it is also called **thermonuclear reactions**.
- **Hydrogen bomb** (mixture of deuterium oxide and tritium dioxide) is the result of nuclear fusion.
- Source of solar (sun) and stellar energy is nuclear fusion.
- The source of emission of large amount of energy during nuclear fission or fusion processes is conversion of mass into energy. It is given by the relation, $E = mc^2$ and calculated in MeV.

Applications of Radioisotopes

Isotopes of all the known elements with $Z > 83$ are radioactive and are called radioisotopes. These are used for various purposes, e.g. radiocarbon dating is used to determine the age of dead specimen with C^{14} content by comparing it with C-12 content.

$$N = N_0 \left(\frac{1}{2}\right)^n$$

where, n = total time$(T)/t_{1/2}$

N_0 = ratio of C^{14}/C^{12} in green plant

N = ratio of C^{14}/C^{12} in wood

Rock dating or uranium dating is used to determine the age of rocks or earth. It is based on $_{82}Pb^{206}$ and $_{92}U^{238}$ ratio.

Uses of other Radioisotopes

Radioisotope	*Uses*
I^{131} (Iodine-131)	(*i*) To study the structure and activity of thyroid gland (*ii*) For the treatment of thyroid disease
I^{123} (Iodine-123)	Brain imaging
Co^{60} (Cobalt-60)	Treatment of cancer
Na^{24} (Sodium-24)	To trace the flow of blood
P^{32} (Phosphorus-32)	For leukemia therapy (blood cancer)
C^{14} (Carbon-14)	To study the kinetics of photosynthesis

CHEMICAL BONDING AND REDOX REACTIONS

CHEMICAL BOND

The forces of attraction like interatomic, intermolecular or interionic forces which holds two or more constituents (atoms, molecules or ions) together are called chemical bonds and the process of their combination is called **chemical bonding**.

In general, it is believed that atoms combine in order to complete their octet, i.e. 8 electrons in their outermost shell (with the exception of hydrogen and helium in case of which outermost shell can have a maximum of two electrons).

Ions

An ion is an electrically charged species. A positively charged ion is called **cation**, e.g. Na^+, H^+, Mg^{2+} while a negatively charged ion is called **anion**, e.g. Cl^-, F^-, I^-.

- A cation contains less electrons than a normal atom while an anion contains more electrons than a normal atom.
- When an atom loses one electron, it carries one positive charge, i.e. converted into cation, and if it gains one electron, it carries one negative charge, i.e. converted into anion.
- There is no change in atomic number or number of protons when an atom forms ion.

Valence Electrons

An electron of an atom, located in the outermost shell (valence shell) of the atom, that can be transferred to or shared with another atom is called valence electrons.

- Elements having same number of valence electrons have similar chemical properties.
- Elements having 1, 2 or 3 valence electrons in their atoms are **metals** and form **cation** while those having 4, 5, 6 or 7 are non-metals and form **anion**.
- Elements with 8 valence electrons (except H and He which have 2 valence electrons) are inert, i.e. unreactive and exist as neutral atom.
- If number of valence electrons = 1, 2 or 3
  Valency = Number of valence electrons
  If number of valence electrons ≥ 4
  Valency = 8 - number of valence electrons

Types of Chemical Bonds

Chemical bonds can be of following types depending upon their mode of formation:

Electrovalent or Ionic Bond

The bond formed by the transfer of electrons from one atom (usually metal) to another (usually non-metal) is called electrovalent bond and the compound is called electrovalent or ionic compound. In other words, it is an intercoulombic forces of attraction between two oppositely charged species. The number of electrons transferred shows the electrovalency of the atom.

The properties of electrovalent compounds are as follow:

- They are usually crystalline solids, i.e. have definite shape and are hard and brittle.
- They have high melting and boiling points.
- They conduct electricity when dissolved in water or in molten state due to the presence of free or mobile ions.
- These are soluble in water and insoluble in organic solvents like alcohol, benzene, etc.
- If the electronegativity difference of two atoms is 1.7, the bond between them is 50% ionic.

Some Electrovalent Compounds

Name	*Formula*	*Ions present*
Aluminium oxide (alumina)	Al_2O_3	Al^{3+} and O^{2-}
Ammonium chloride	NH_4Cl	NH_4^+ and Cl^-
Copper sulphate	$CuSO_4$	Cu^{2+} and SO_4^{2-}
Magnesium chloride	$MgCl_2$	Mg^{2+} and Cl^-
Magnesium oxide	MgO	Mg^{2+} and O^{2-}
Sodium chloride	$NaCl$	Na^+ and Cl^-
Sodium hydroxide	$NaOH$	Na^+ and OH^-

Covalent Bond

The bond formed by the sharing of electrons between two atoms of same or different elements is called **covalent bond** and the compound is called **covalent compound**. When a non-metal combines with another non-metal, a covalent bond is formed. Covalent bond may be single, double or triple depending upon the number of sharing pairs of electrons. The number of electrons shared shows the covalency of the element.

The properties of covalent compounds are as follow:

- Normal covalent compounds are gases or liquid under ordinary conditions of temperature and pressure.
- These do not conduct electricity (due to the absence of free ions) and are insoluble in water but dissolve in organic solvents.
- Graphite although is covalent but conducts electricity because of the presence of free electrons (electrical conduction).
- These show stereoisomerism because covalent bond is directional in nature.

Some Covalent Compounds

Name	*Formula*	*Elements present*
Alcohol (ethanol)	$C_2H_5(OH)$	C, H and O
Ammonia	NH_3	N and H
Acetylene (ethyne)	C_2H_2	C and H
Carbon disulphide	CS_2	C and S
Carbon tetrachloride	CCl_4	C and Cl
Cane sugar	$C_{12}H_{22}O_{11}$	C, H and O
Ethylene	C_2H_4	C and H
Glucose	$C_6H_{12}O_6$	C, H and O

Coordinate or Dative Bond

This type of bond is formed by one sided sharing of one pair of electrons between two atoms. The atom having complete octet which provides the electron pair for sharing is known as donor atom. The other atom which accepts the electron pair is called the **acceptor atom**,

e.g. $NH_3 \longrightarrow BF_3, NH_4^+$

Hydrogen Bond

This bond is formed due to the electrostatic force of attraction between hydrogen atom highly electronegative atom.

i.e. N, O or F and any other electronegative atom which is present in the same or different molecules.

It is mainly of two types:

(i) **Intermolecular H-bonding** (e.g. H_2O, HF, NH_3 molecule). It occurs between different molecules of a compound.

(ii) **Intramolecular H-bonding** (e.g. *o*-nitrophenol)

It occurs with in different parts of a same molecule.

- Molecules having O—H, N—H or H—F bond show abnormal properties due to H—bond formation.

Redox Reactions

The reactions which involve both oxidation and reduction as its two half reactions are called **redox reactions**. When the same element is oxidised or reduced simultaneously, the reaction is called disproportionation reaction.

Oxidation

The process which involves gain of oxygen or any other electronegative element or loss of hydrogen or loss of one or more electrons (de-electronation) from an atom, ion or molecule is called oxidation,

e.g. $Mg \longrightarrow Mg^{2+} + 2e^-$.

The positive valency of an element increases by its oxidation.

> **RANCIDITY**
>
> The oils or fats containing materials get oxidised in the presence of oxygen or air and produce foul odour and taste. This process is called rancidity. It is the source of destructive free radicals. A sliced apple turns brown if kept open for sometimes due to oxidation of iron present in the apple. The chips packets are filled with inert gas nitrogen to protect them from being oxidised (or rancid).

Reduction

The process which involves the gain of hydrogen or electropositive element or one or more electrons (electronation) or loss of oxygen by an atom, ion or molecule is called reduction.

e.g. $S + 2e^- \longrightarrow S^{2-}$.

Reduction involves decrease in the positive valency of an element.

Oxidising Agent (Oxidant)

It is a substance which accepts electron in the chemical reaction, i.e. electron acceptors are oxidising agent. All the positively charged species behave like oxidising agents. Oxidising agents are Lewis acids.

Reducing Agent (Reductant)

The substance which donates electron in a chemical reaction is called reducing agent, i.e. electron donors are reducing agents. All the negatively charged species behave like reducing agents. Reducing agents are Lewis bases.

Some Oxidising and Reducing Agents

Oxidising agents	*Reducing agents*	*Both oxidising and reducing agents*
$KMnO_4$, $Cr_2O_7^{2-}$, H_2SO_4, HNO_3, O_2, CO_2 etc.	Na, Al, Fe, Zn, LiH, NaH, $(COOH)_2$ etc.	H_2O_2, SO_2, HNO_2, $NaNO_2$, O_3, Na_2SO_3 etc.

Oxidation State

It is the real or imaginary charge which an atom appears to have in its combined state. Oxidation state of an element may be positive, negative, zero or fractional.

Rules for Determining Oxidation State

- The oxidation state of an element in its free or uncombined state is zero. e.g. oxidation state of O in O_2 and O_3 is zero.
- Oxidation state of hydrogen in most of its compounds is +1 but in metallic hydrides it is –1.
- Oxidation state of oxygen in most of its compounds is – 2. Except peroxides (–1), superoxides (–1/2) and oxygen fluorides (+2 or +1).

- Oxidation state of elements of IA, IIA and IIIA subgroups in their compounds are + 1, + 2 and + 3, respectively.
- Oxidation state of any ion is equal to its charge present on it.
- The algebraic sum of oxidation states of all the elements in the neutral molecule is zero.
- The algebraic sum of the oxidation states of all elements present in polyatomic ion is equal to the charge on the ion.
- Oxidation state of fluorine (F) is always –1.

➠ **Note** Compounds of all elements with oxygen are called oxides with the exception of fluorine in case of which it is called fluorides, e.g. oxygen fluoride (OF_2).

GAS LAWS AND SOLUTIONS

GAS LAWS

Gases exhibit dependency on temperature, pressure, volume and mass. These interrelations can be analysed by gas laws.

At STP/NTP, temperature, $(T) = 273.15$ K, pressure $(p) = 1$ atm $= 101.35$ kPa, volume, $(V) = 22.4$ L mol^{-1}

Some gas laws are given below

Boyle's Law (Robert Boyle-1662)

- At constant temperature, the volume of fixed mass of a gas is inversely proportional to its pressure.

$$V \propto \frac{1}{p} \quad \text{[at constant temperature and moles } (n)]$$

or $\quad p_1V_1 = p_2V_2$

Charles' Law (Jacques Charles-1787)

- At constant pressure, volume of fixed mass of a gas is directly proportional to the absolute temperature.

$$V \propto T \quad \text{(at constant pressure and moles)}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Gay Lussac's Law (J.Gay Lussac-1802)

- At constant volume, the pressure of fixed mass of a gas is directly proportional to its temperature.

$$p \propto T$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2} \quad \text{[at constant volume and moles]}$$

Avogadro's Law (Amedeo Avogadro-1811)

- At constant temperature and pressure, the volume of any gas is directly proportional to the number of moles of gas.

$$V \propto n \quad \text{[at constant } T \text{ and } P]$$

or $\quad \dfrac{V_1}{n_1} = \dfrac{V_2}{n_2}$

GAS EQUATION

The three laws (Boyle's, Charle's and Avogadro's law) can be combined together in a single equation which is known as ideal gas equation.

$$V \propto 1/p \quad \text{(Boyle's law)}$$

$$V \propto T \quad \text{(Charles' law)}$$

$$V \propto n \quad \text{(Avogadro's law)}$$

On combining, $V \propto \dfrac{nT}{p}$

or $\quad V = R\dfrac{nT}{P}$

or $\quad pV = nRT \quad$ (where, R = gas constant)

For 1 mole, $\quad pV = RT$

- Value of gas constant, R depends upon the units of measurements.

$$R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$$
$$= 8.3143 \text{ J mol}^{-1} \text{ K}^{-1} = 5.189 \times 10^{19} \text{eV mol}^{-1} \text{ K}^{-1}$$
$$= 1.99 \text{ cal mol}^{-1} \text{ K}^{-1}$$

Gas constant, R for a single molecule is called Boltzmann constant (k) or $k = \dfrac{R}{N} = \dfrac{R}{6.022 \times 10^{23}}$

- Density of a gas is directly proportional to pressure and inversely related to temperature,

i.e. $\quad d = pM/RT \quad \left[\because n = \dfrac{Mass\ (m)}{Molecular\ weight\ (M)}\right]$

Graham's Law of Diffusion (or Effusion)

The rate of diffusion (r) of a gas at constant temperature and pressure is inversely proportional to the square root of its molecular mass, M or density, d.

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{d_2}{d_1}}$$

Maxwell's Distribution of Molecular Speeds (Velocities)

Maxwell and Boltzmann proposed that gas molecules are always in rapid random motion colliding with each other and with the walls of container because of which their velocity changes. On increasing temperature, the velocity or molecular motion increases because of which the rate of reaction increases.

SOLUTION

A solution is a homogeneous mixture of two or more substances on molecular level is called true solution or solution. e.g. lemonade, soda water. The substance which is dissolved in a liquid to make a solution is called 'solute' and the liquid in which the solute is dissolved is called 'solvent'.

A true solution does not scatter light and its particles cannot be seen even by microscope. e.g. salt solution, sea water, sugar solution, copper sulphate solution, vinegar, etc.

Depending upon the amount of solute present, the solutions can be classified as :

- A solution, in which more quantity of solute can be dissolved without raising its temperature is called an **unsaturated solution**.
- A solution, in which no more solute can be dissolved at that temperature is called a **saturated solution**.

Concentration of a Solution

The amount of a solute dissolved in unit weight or volume of solution is called strength (concentration) of a solution.

So, concentration of a solution

$$= \frac{\text{Amount of solute (in g)}}{\text{Weight of solution}} = \frac{\text{Amount of solute (in g)}}{\text{Volume of solution}}$$

Different Terms to Express Concentration

Parts Per Million It shows the parts of solute present per million parts of solution, i.e. ppm.

$$= \frac{\text{Mass of solute (in g)} \times 10^6}{\text{Volume of solution (in mL)}}$$

Molarity (M) The number of moles of solute dissolved in one litre of solution is called its molarity.

Thus, $$M = \frac{\text{Weight of solute (in g)} \times 1000}{\text{Molecular weight} \times \text{volume of solution (in mL)}}$$

Molality (m) The number of moles of solute dissolved in 1000 g of a solvent is called its molality.

Thus, $$m = \frac{\text{Weight of solute (in g)} \times 1000}{\text{Molecular weight} \times \text{weight of solvent (in g)}}$$

Normality (N) The number of gram equivalents of solute dissolved in one litre of solution is known as its normality.

$$N = \frac{\text{Weight of solute (in g)} \times 1000}{\text{Equivalent weight} \times \text{volume of solution (in mL)}}$$

➨ **Note** Normality and molarity are affected by temperature as these depend upon the volume whereas molality remains unaffected from temperature change.

Henry's Law

According to this law, 'at particular temperature, the solubility of a gas in a liquid is directly proportional to the pressure of the gas above the solvent.

Its main applications are as follow:

- Soft drinks and soda water bottles are sealed under high pressure in order to increase the solubility of CO_2 in them.
- To minimise the painful effects (bends) accompanying the decompression of deep sea divers, oxygen diluted with less soluble helium gas is used as breathing gas.
- Oxygen diluted with nitrogen cannot be used for this purpose due to high solubility of N_2 in blood.

Osmosis

It is the process of movement of solvent molecules from the solution of low concentration to high concentration through semipermeable membrane.

- If pressure greater than osmotic pressure (pressure require to stop osmosis) is applied on the solution of high concentration, **reverse osmosis** takes place. e.g. desalination of sea water.
- **Isotonic solutions** have same concentration and osmotic pressure. e.g. 0.91% solution of pure NaCl is isotonic with human red blood cells (RBCs).

Diffusion

- Diffusion is the process of spontaneous mixing of different gases and in this process, molecules of a substance move from higher concentration to lower concentration and goes on until a uniform mixture is formed. Its rate is highest for gases and lowest in solids. e.g. the smell of food being cooked, reaches us even from a considerable distance by the process of diffusion. Due to diffusion the scent spread all over the room if the lid is opened.